



Introduction to Deep Learning

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AI342

With materials from Fuxin Lee, and Alexander Amini

What is deep Learning?

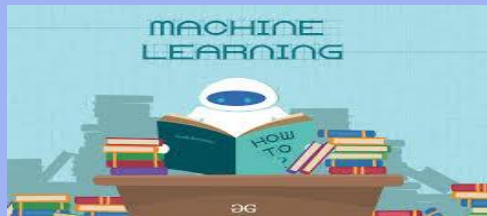
Artificial Intelligence

Any technique that enable the machine to learn and reason like human.



Machine Learning

Techniques that train the machine to recognize patterns without being explicitly programmed



Deep Learning

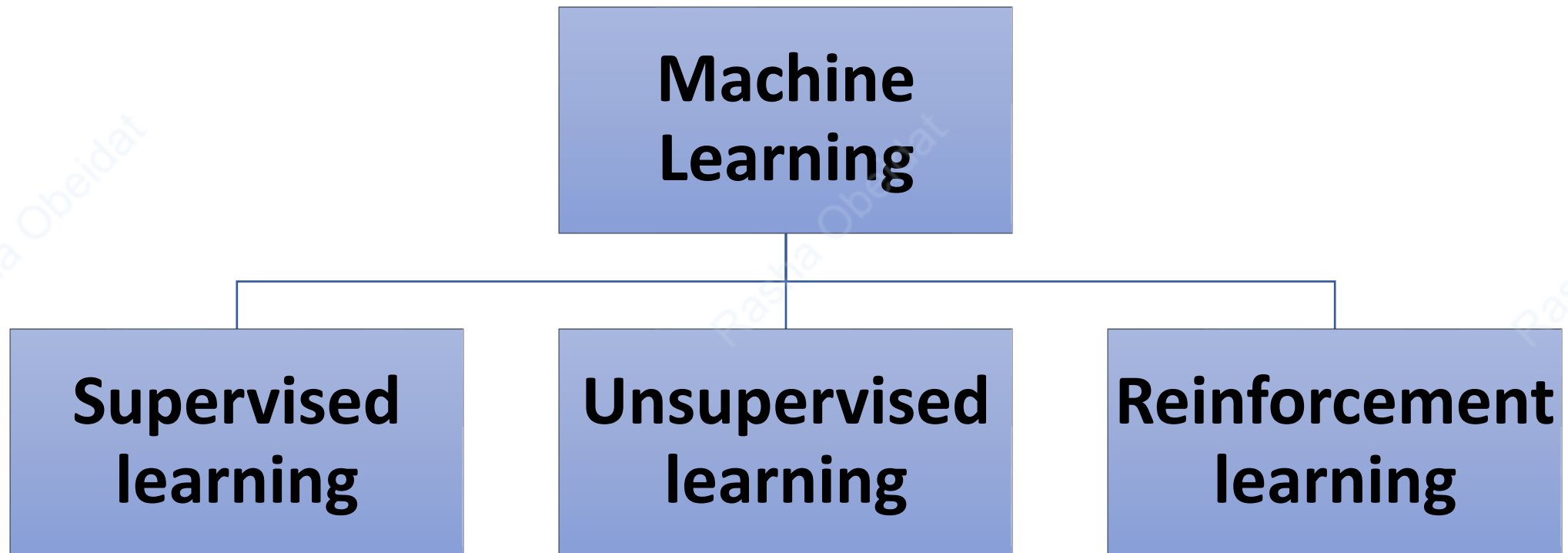
Technique that automatically recognize patterns from the data using deep neural networks



Machine Learning

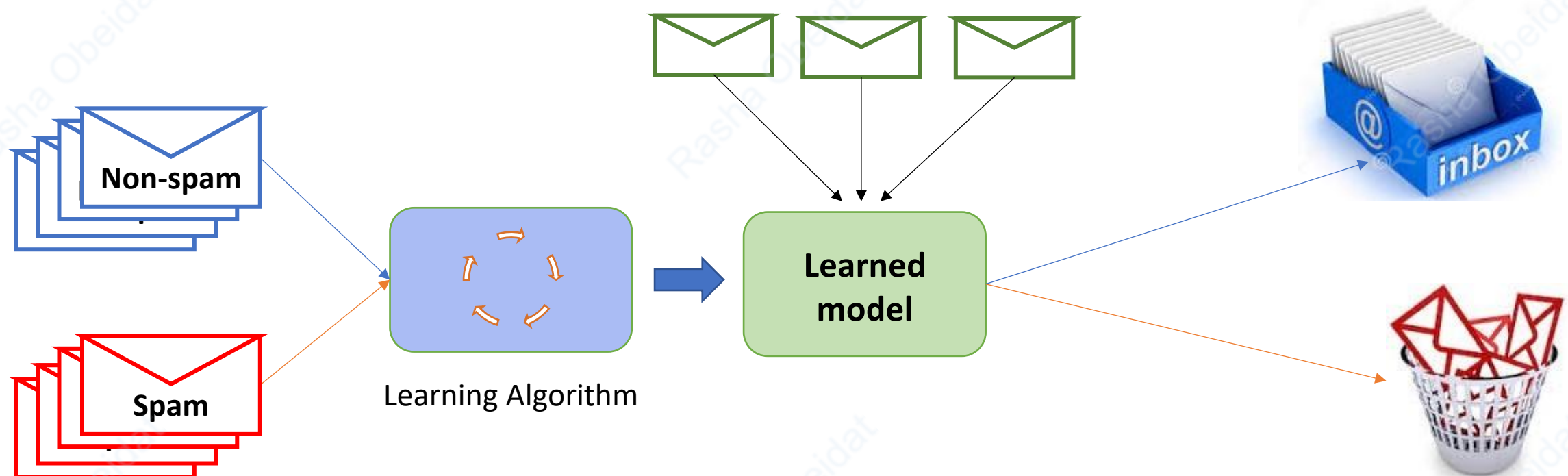
- **Machine learning focuses on developing computer programs** that can access data and use it automatically learn patterns.
- Data consists of a set of examples. The patterns learned from the available examples can be used later to make decisions on new examples that are going to be provided in the future.
- **Note:** Data examples are also known as samples or instances.

Learning Types in Machine Learning.



Supervised Learning

- Set of algorithms that enables that machine to learn how to classify examples into one or more of a **predefined set of labels**.
 - labels also known as classes, categories, and types.



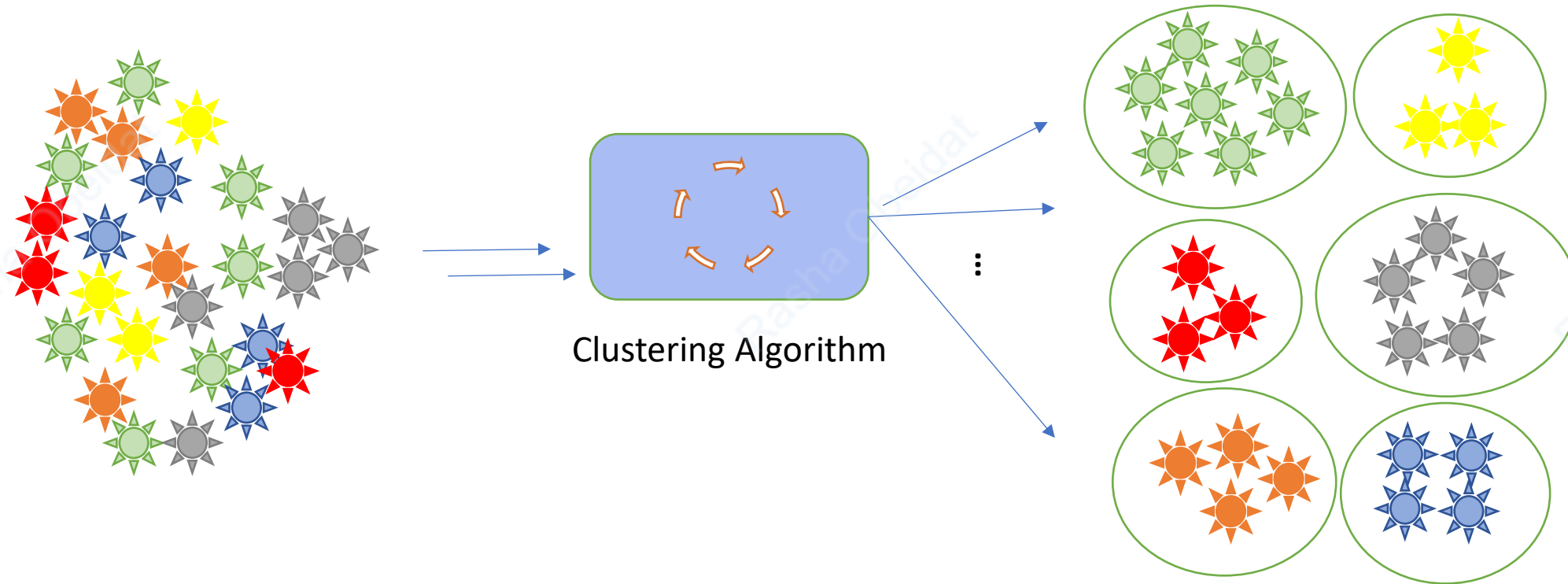
Supervised Learning

- **Input:** data in the form of examples-labels pairs.
- **Training:** a learning algorithm is fed with these example-label pairs. The algorithm to predict a label for each example (that already has a ground truth label or correct label), feedback of whether it predicted label matches the correct label is used to **adjust** the learned model. eventually, the algorithm learns to approximate the relationship between examples and their labels.
- **Testing:** the fully-trained model will be used later to predict a label for a new **unlabeled example** that has never seen before.

Unsupervised Learning

- **Input:** data in the form of data points with no corresponding labels.
- The algorithm learns to ***cluster (group)*** the data points such that the similar points placed in the same cluster.
- Unsupervised learning is important because the majority of the available data is **not labeled**.
- **Semi-supervised learning** algorithms are hybrid learning techniques that learn from both supervised and unsupervised data.

Unsupervised Learning



Reinforcement Learning

- Reinforcement Learning can be defined as learning from mistakes. It is An **agent** that Observes the **environment**, takes **action**, and receives **rewards/punishments** for each action.
- We reinforce our agents to prefer good behaviors over bad ones. In the beginning, the agent makes a lot of mistakes. Over time, the agent learns to make fewer mistakes than it used to do.

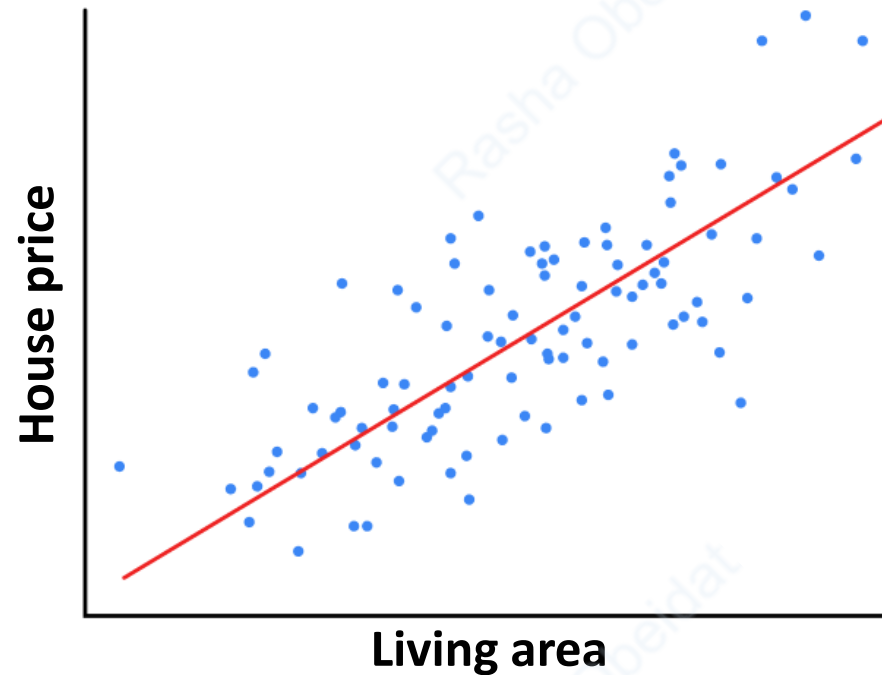


Supervised Learning: basics.

- Turn every example into a vector $x_i \in \mathbb{R}^d$.
- Training set consists of a set of example-label pairs: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n) = \{(x_i, y_i)\}_{i=1}^n$
- y_i is corresponding output of the x_i .
 - y_i could be *continuous* (as in regression)
 - *discrete* (as in classification).
- We assume that there is some underlying function $f(x)$ that maps x to y .
- Our goal is to find an approximation $g(x)$ of the function $f(x)$.

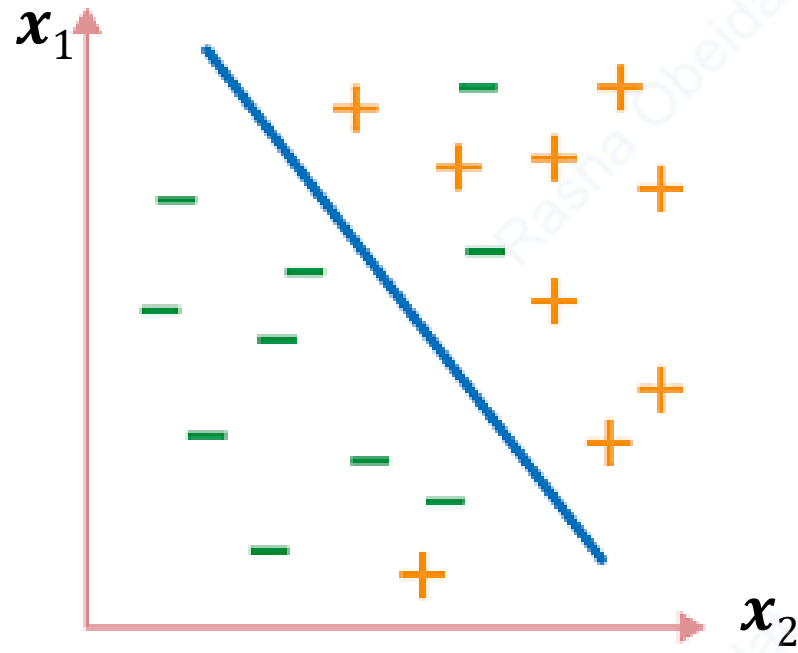
Regression

- The space of the output is continuous.
- E.g., estimating the price of the house perhaps in the range of JD50,000 to JD150,000 based on the living area.



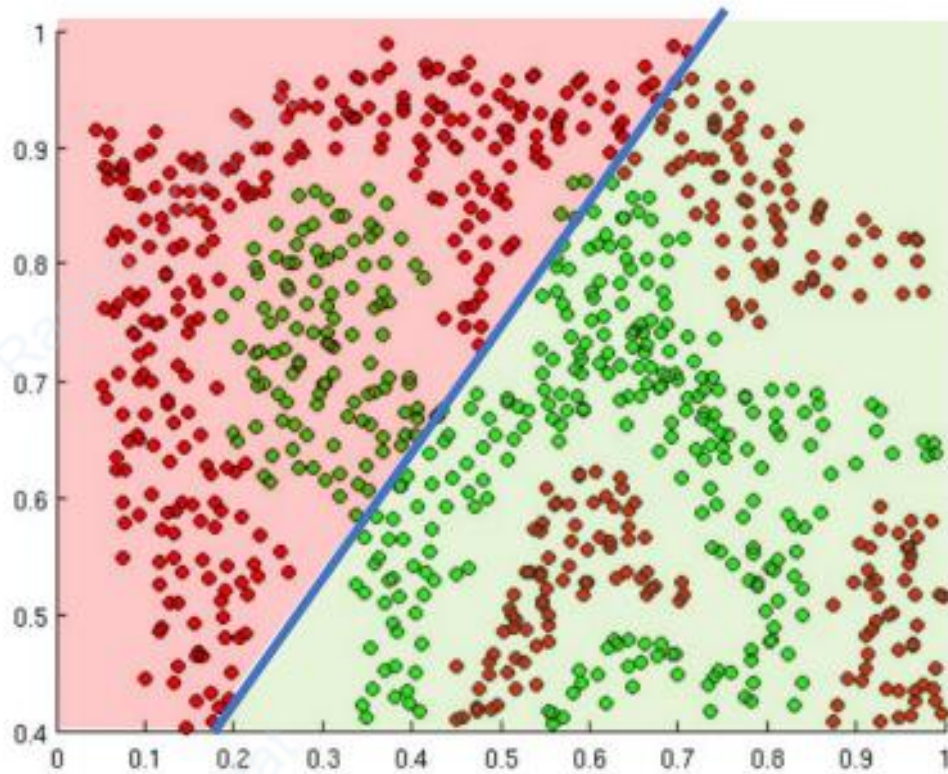
Classification

- The space of the output is discrete (uses a predefined set of classes).
- Classify an email to a spam or non-spam.

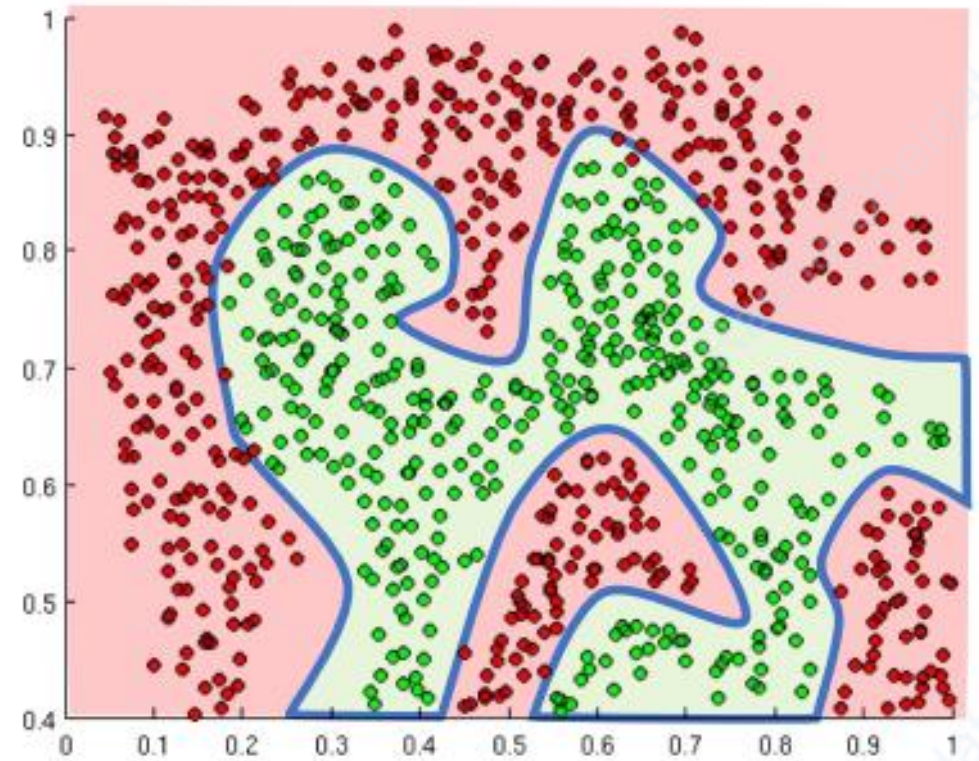


Linear Classifier vs Nonlinear classifier

- Classifiers can be broadly categorized into **linear** and **nonlinear** classifiers based on their decision boundaries.



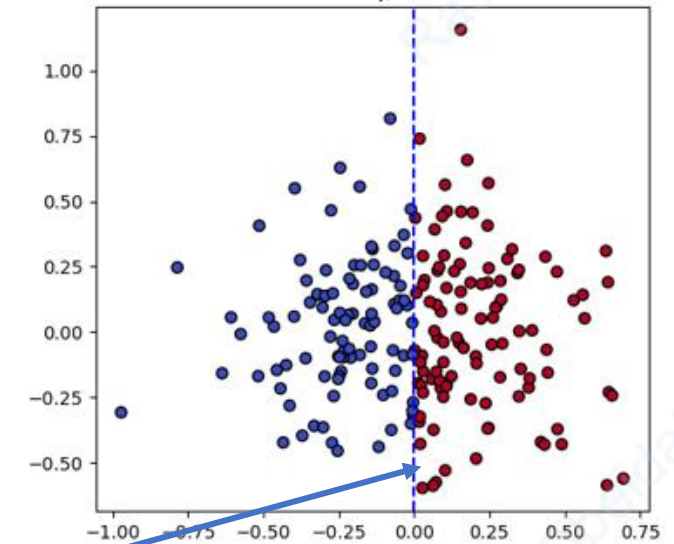
linear classifier



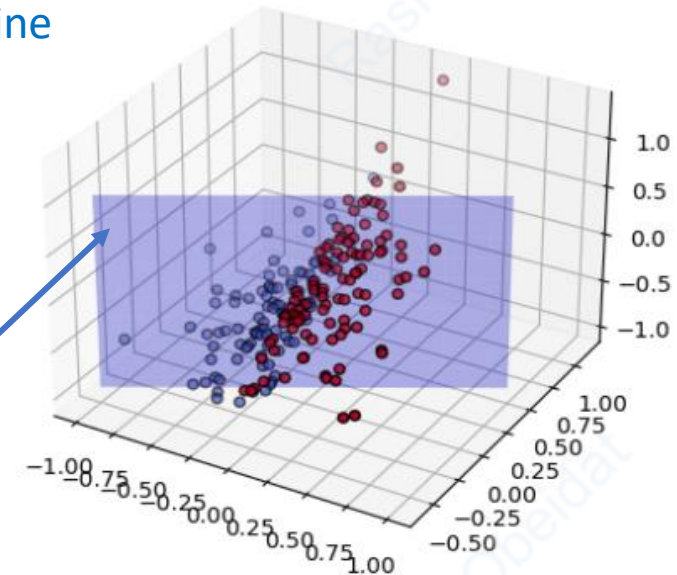
nonlinear
classifier

Linear Classifier

- **Definition:** A linear classifier separates data using a straight line (or a hyperplane in higher dimensions).
- **Underlying assumption:** It assumes that the data is linearly separable.
- **Examples:** Logistic Regression, Support Vector Machines (with linear kernels), and Perceptron.
- **Advantage:** Computationally efficient and interpretable.
- **Limitation:** Not effective for complex data distributions.



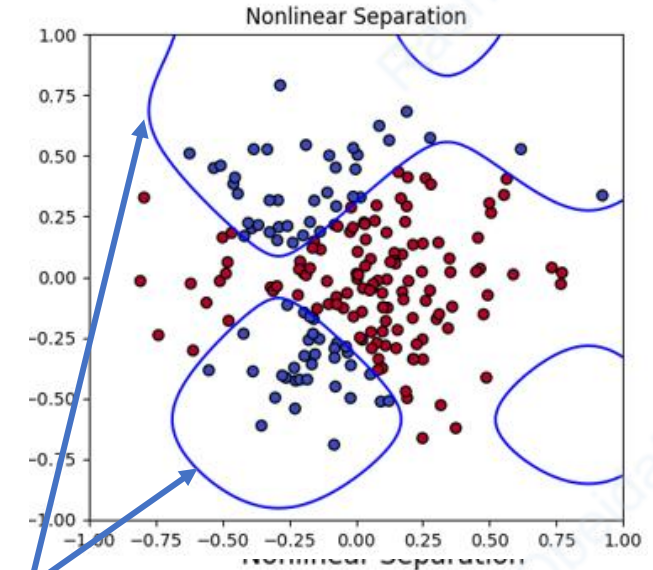
Linear
Separation line



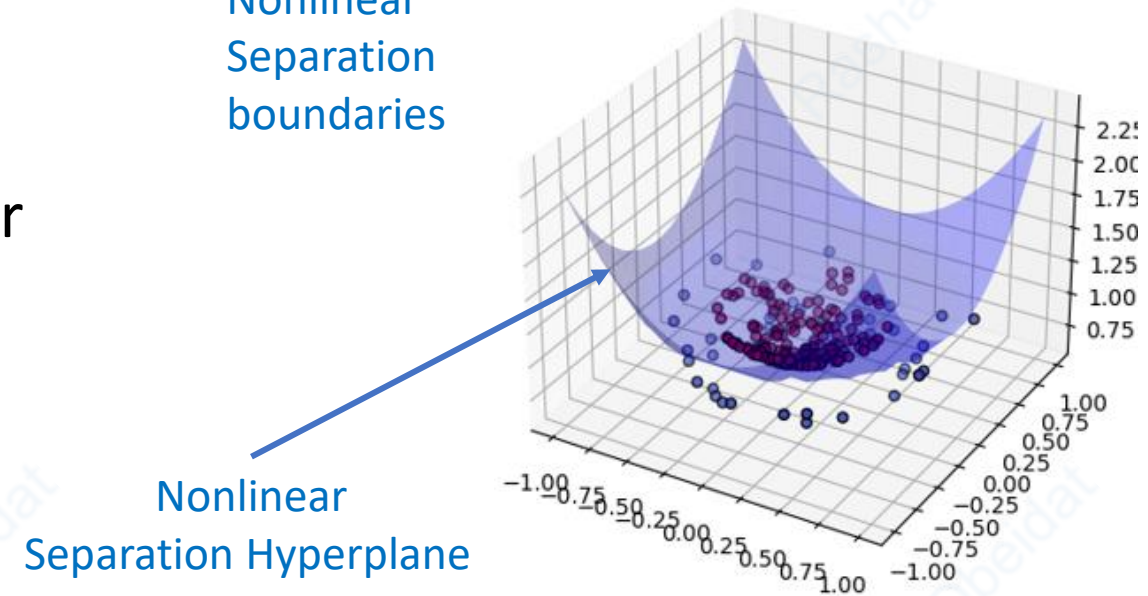
Linear separation
Hyperplane

Nonlinear Classifier

- **Definition:** A nonlinear classifier uses more complex decision boundaries to separate data.
- **Underlying assumption:** It assumes that the data is nonlinearly separable.
- **Examples:** Decision Trees, Support Vector Machines (with nonlinear kernels), Neural Networks.
- **Advantage:** More flexible and powerful for handling complex data.
- **Limitation:** May require more data and computational resources.



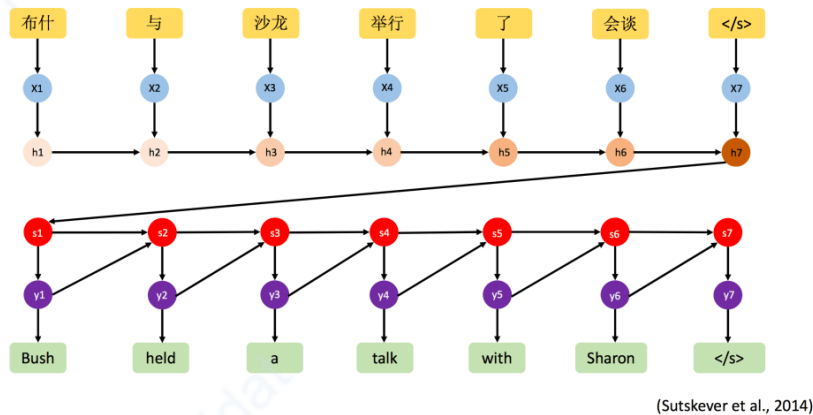
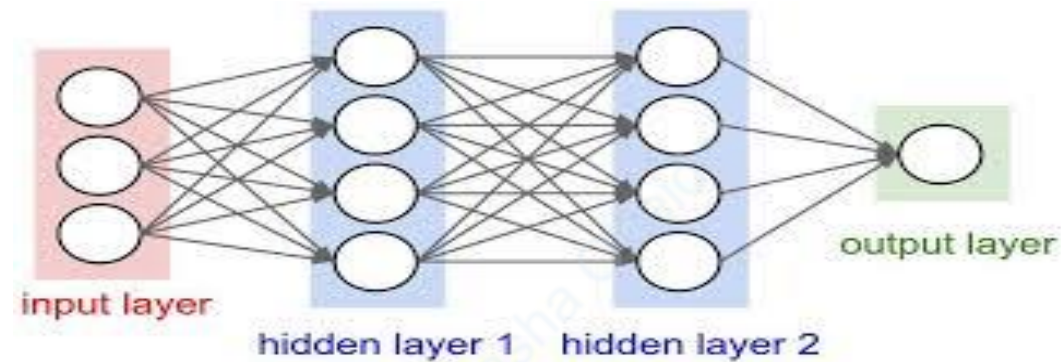
Nonlinear
Separation
boundaries



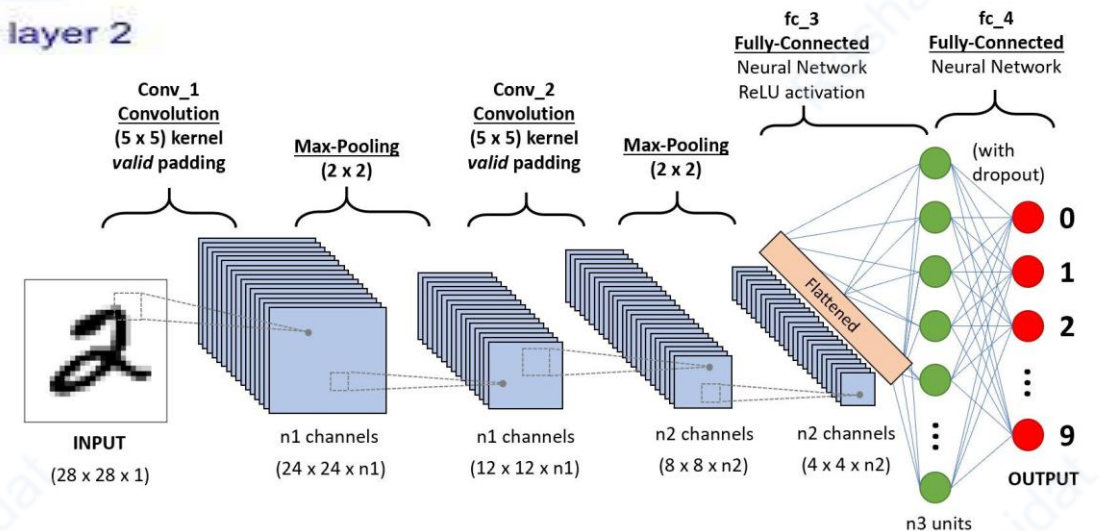
Nonlinear
Separation Hyperplane

Deep learning (DL)

- An area of **Machine Learning** that uses deep learning architectures to recognize pattern, learning can be supervised, unsupervised or semi-supervised.

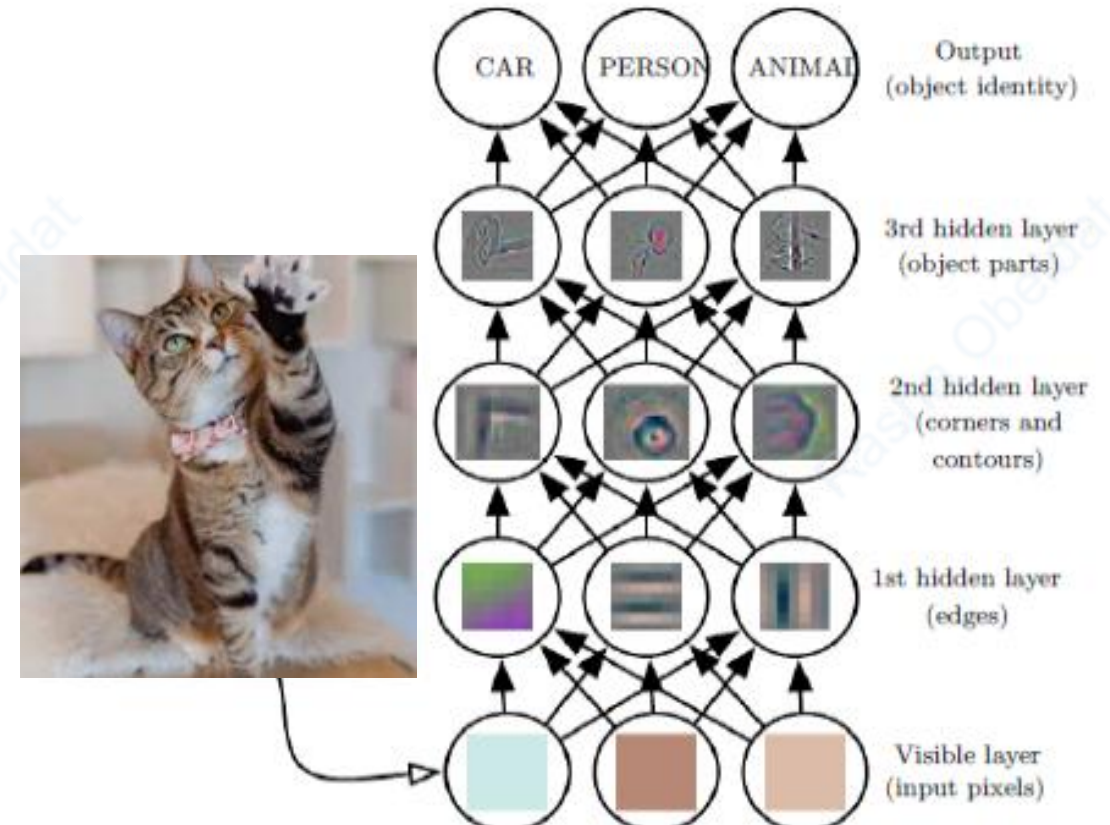


(Sutskever et al., 2014)



How Deep Learning Understands the World

- Deep learning represents the world as a nested ***hierarchy of concepts***, with each concept defined in relation to simpler concepts.
- It involve a greater amount of composition of learned functions or learned concepts than traditional machine learning does (it is the learning of *multiple levels of composition*).



DL: What Makes It So Powerful?

1. Eliminates Manual Feature Engineering
2. Neural Networks are Universal Approximators

Hand-Feature Engineering

- **Feature engineering** is the process of **transforming raw data into meaningful input features** that improve the performance of machine learning models.
- Feature engineering Essential for **Traditional ML** – Unlike deep learning, most ML models **do not** learn features automatically.

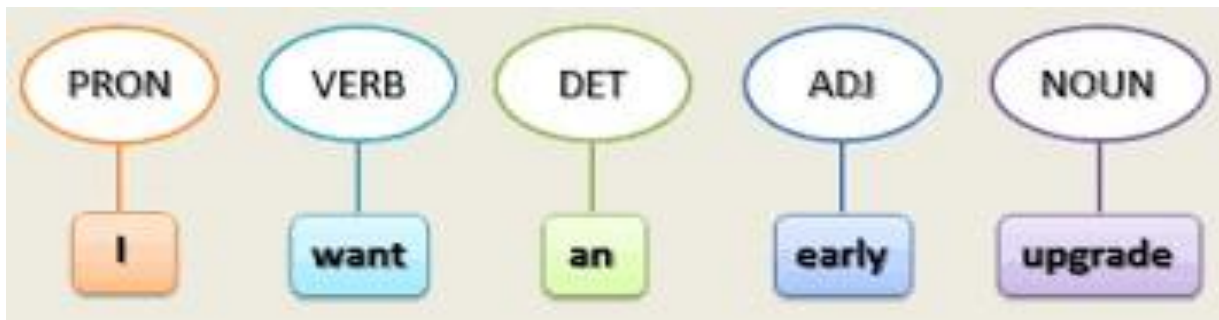
Hand-Feature Engineering

- **Limitations :**

- **Requires Domain Experts** – In many cases, deep knowledge of the domain is necessary to create meaningful features.
- **Time-Consuming** – Requires domain expertise and extensive trial-and-error
- **Scalability Issues** – Manually engineered features may not generalize well to new data.
- **Bias & Human Error** – Engineers may introduce subjective biases when selecting features.
- **Difficult for Complex Data** – Traditional feature engineering struggles with high-dimensional, unstructured data like images and text.

Feature Engineering in DL

- Deep Learning **learns feature Automatically**, reducing the need for manual work.
- **Example: POS tagging.**
 - It is the process of marking up a word in a text (corpus) as corresponding to a particular part of speech, based on both its definition and its context.



Example Features:

The word shape: Aaa, aaa, N-grams.

Previous word.

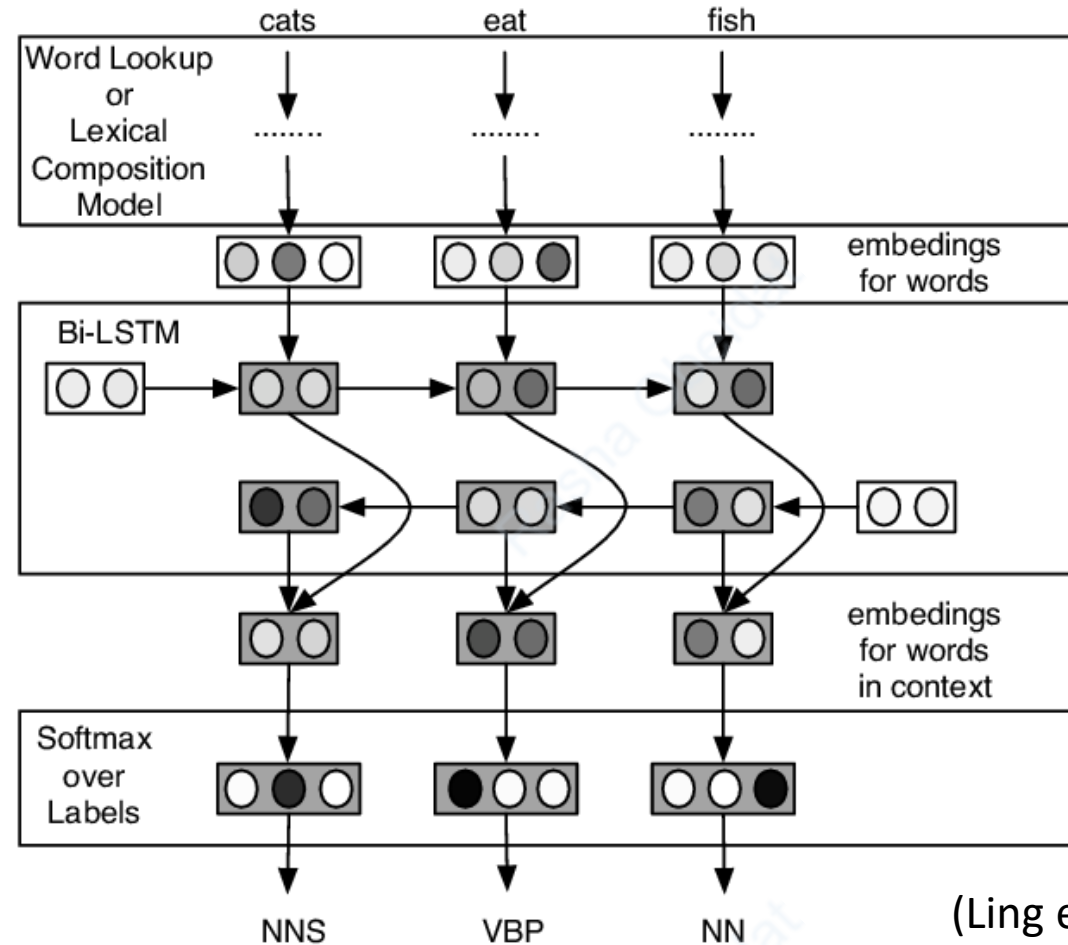
Next word.

POS tag of Previous word.

POS tag of next word.

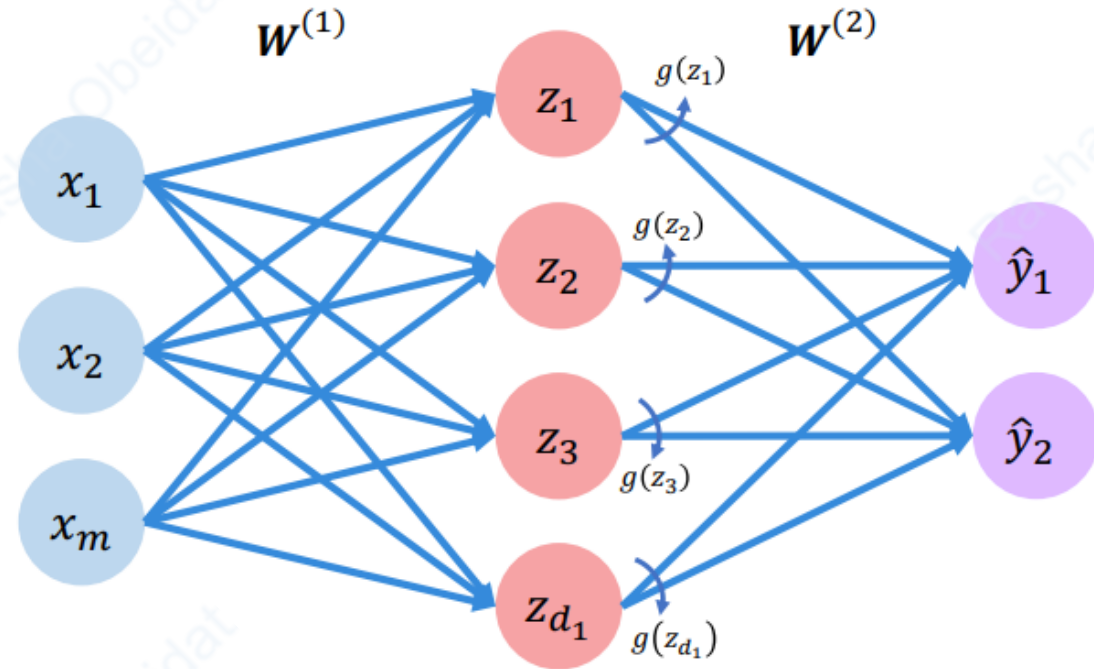
Word cluster.

Example: POS Tagging using Deep Learning



Neural networks are universal Approximators

- A feed-forward network with a single hidden layer containing a finite number of neurons can approximate any continuous functions up to arbitrary precision (ϵ).

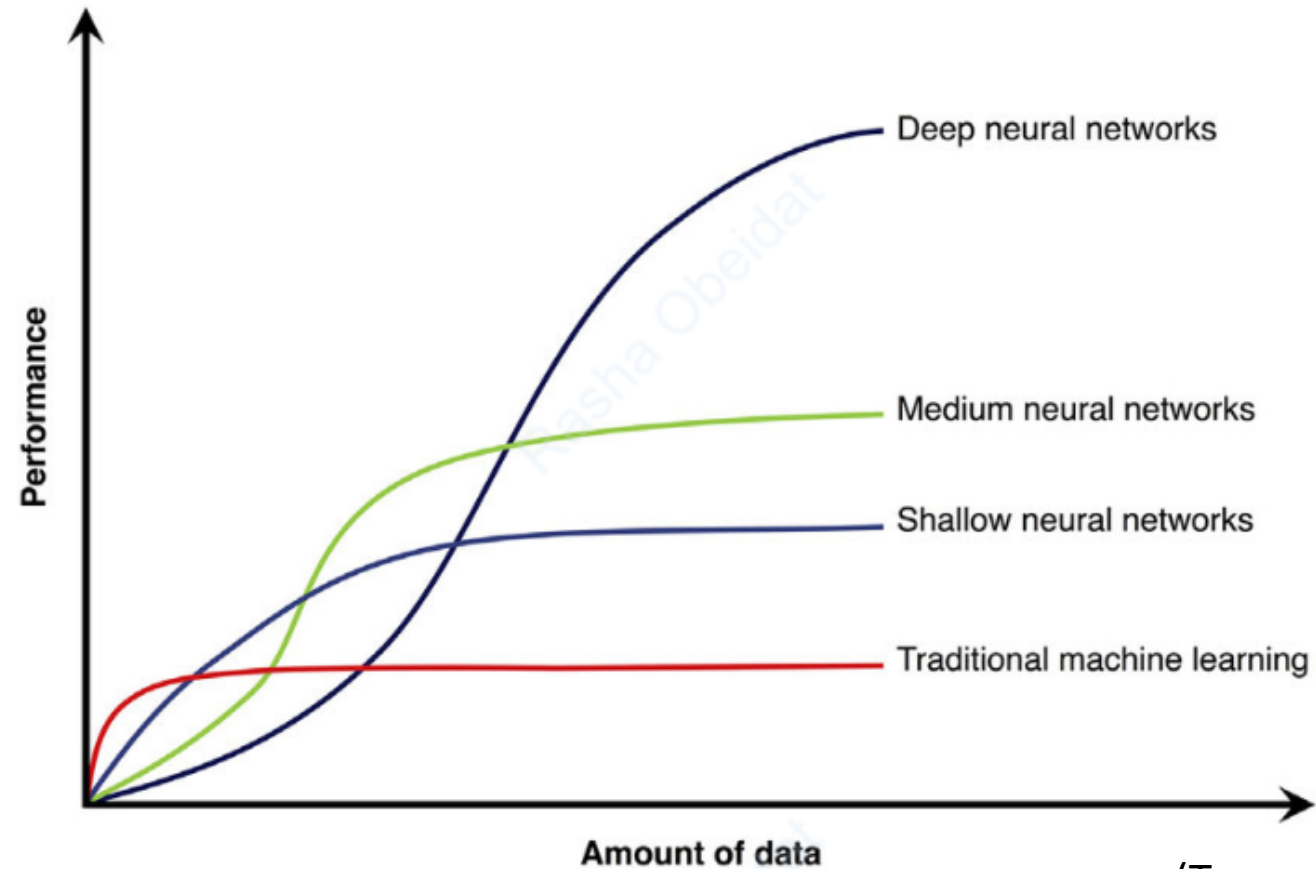


Deep learning: Why now?

- **Availability of data** has grown exponentially. E.g., Social media, digital photos, sensors, smart phones, etc.
- **Improvement in computer hardware and software infrastructure.** GPU are heavily used to train deep model because it is much faster than CPUs. GPU provides high-bandwidth memories and hardware that performs floating-point arithmetic at significantly higher rates than conventional CPUs.



The impact of Data Availability on deep learning.



Application

- **Computer Science**
 - **Natural Language Processing** :machine translation, sentiment analysis, information extraction, KB population, parsing, etc.
 - **Computer Vision**: image classification, object detection, image retrieval, image caption generation.
 - **Speech recognition**.
- **Biology**: Identifying gene coding regions in a genome, protein structure prediction.
- **Chemistry**: Prediction of Chemical Patterns,
- **Urban Planning**: Urban growth and land-use change prediction
- **Pharma and Medicine**: Personalized Treatment, Drug Discovery, Epidemic Outbreak Prediction.

Example: Recognizing Handwritten Digits

- MNIST database
 - 60,000 training, 10,000 testing
 - Large enough for digits
 - Battlefield of the 90s



Algorithm	Error Rate (%)
Linear classifier (perceptron)	12.0
K-nearest-neighbors	5.0
Boosting	1.26
SVM	1.4
Neural Network	1.6
Convolutional Neural Networks	0.95
With automatic distortions + ensemble + many tricks	0.23

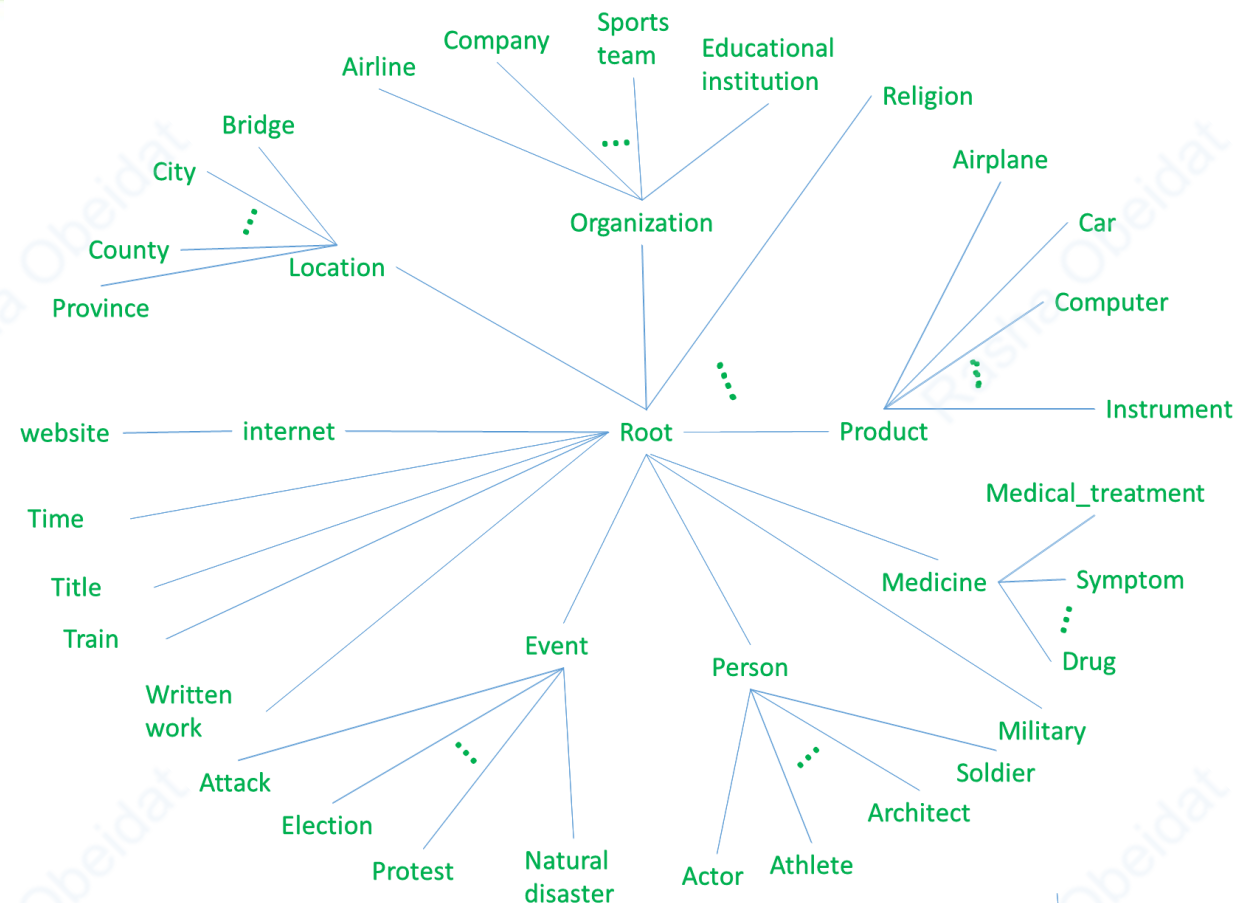
Using deep neural networks

Example: Fine Grained Entity Typing (FGET).

heroes, [Mr. McCain]_{person/politician} carried his renowned name into battle and into political fights for more than a half-century. It was an odyssey

A son and grandson of four-star admirals who were his larger-than-life heroes, [Mr. McCain] carried his renowned name into battle and into political fights for more than a half-century. It was an odyssey driven by raw ambition, the conservative instincts of a shrewd military man, a rebelliousness evident since childhood and a temper that sometimes bordered on explosiveness.

- NER: small set of coarse types (e.g. per, Loc, org)
- larger set of types.
- Type hierarchy
- Multi-label classification



Example: Fine grained entity typing

Typing methods	Wiki/FIGER(GOLD)			OntoNotes			BBN		
	Acc.	Ma-F1	Mi-F1	Acc.	Ma-F1	Mi-F1	Acc.	Ma-F1	Mi-F1
FIGER [*] (Ling and Weld, 2012)	0.474	0.692	0.655	0.369	0.578	0.516	0.467	0.672	0.612
HYENA [*] (Yosef et al., 2012)	0.288	0.528	0.506	0.249	0.497	0.446	0.523	0.576	0.587
AFET-NoCo [*] (Ren et al., 2016)	0.526	0.693	0.654	0.486	0.652	0.594	0.655	0.711	0.716
AFET-CoH [*] (Ren et al., 2016)	0.433	0.583	0.551	0.521	0.680	0.609	0.657	0.703	0.712
AFET [*] (Ren et al., 2016)	0.533	0.693	0.664	0.551	0.711	0.647	0.670	0.727	0.735
AFET ^{†‡} (Ren et al., 2016)	0.509	0.689	0.653	0.553	0.712	0.646	0.683	0.744	0.747
Attentive [†] (Shimaoka et al., 2016)	0.581	0.780	0.744	0.473	0.655	0.586	0.484	0.732	0.724
AAA-AIIC [†]	0.662	0.805	0.770	0.514	0.672	0.626	0.655	0.736	0.752
AAA-NoM [†]	0.646	0.808	0.768	0.521	0.683	0.626	0.615	0.742	0.755
AAA	0.658	0.812	0.774	0.522	0.685	0.633	0.604	0.741	0.757



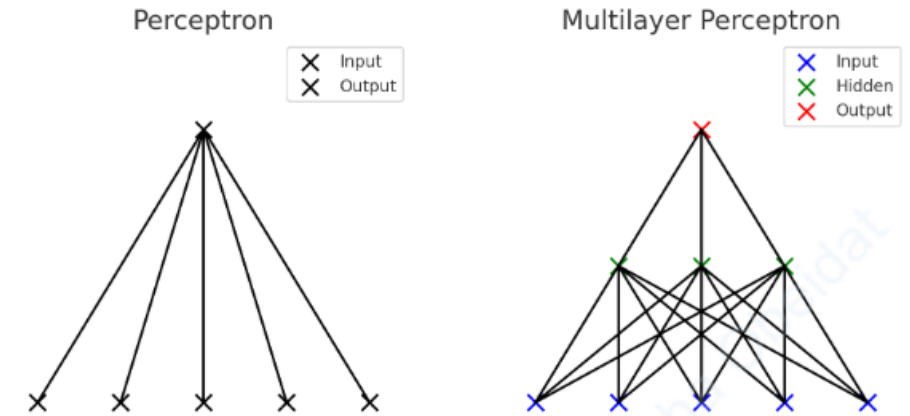
(Abhishek et. Al 2017)

Using deep neural networks

Key Milestones in Deep Learning History

1940s-1950s: Early Theoretical Foundations

1. 1943: Warren McCulloch and Walter Pitts introduced the first artificial neuron model.
2. 1958: Frank Rosenblatt developed the **Perceptron**, an early form of a neural network capable of binary classification.



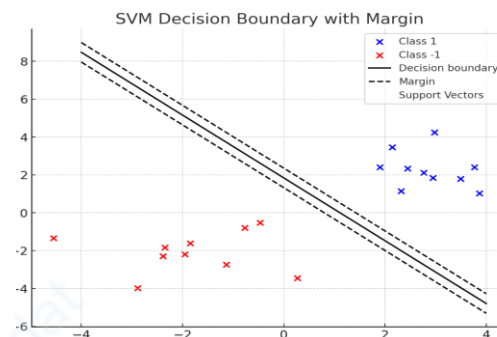
1960s-1980s: Challenges and Neural Network Stagnation

1. 1969: Marvin Minsky and Seymour Papert's book *Perceptrons* highlighted the limitations of single-layer perceptrons, leading to reduced interest in neural networks.
2. 1986: Geoffrey Hinton, David Rumelhart, and Ronald Williams introduced the **Backpropagation Algorithm**, enabling training of multi-layer networks.

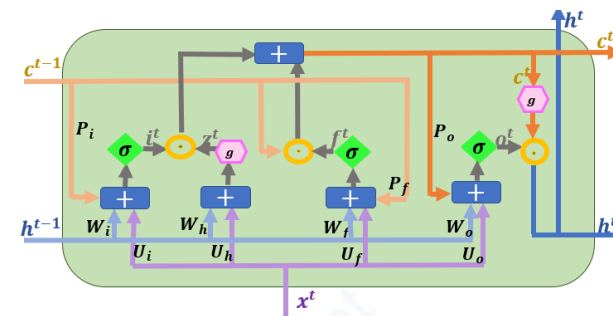
Key Milestones in Deep Learning History

1990s-2000s: Foundations of Modern Deep Learning

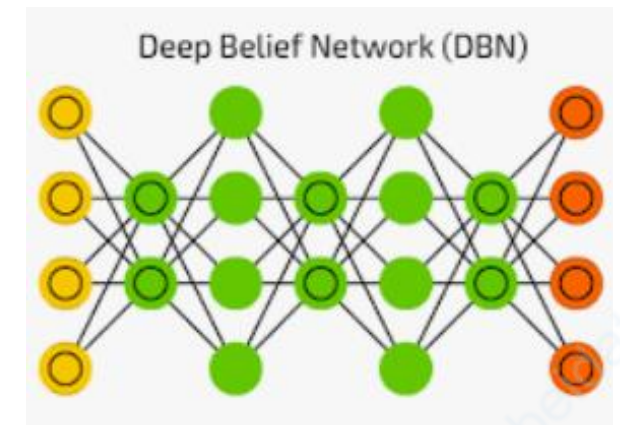
1. 1995: Vapnik introduced **Support Vector Machines (SVMs)**, which competed with neural networks.
2. 1997: Sepp Hochreiter and Jürgen Schmidhuber developed **Long Short-Term Memory (LSTM)**, a breakthrough for sequence learning.
3. 2006: Geoffrey Hinton and colleagues introduced **Deep Belief Networks (DBNs)**, rekindling interest in deep learning.



Support Vector Machines



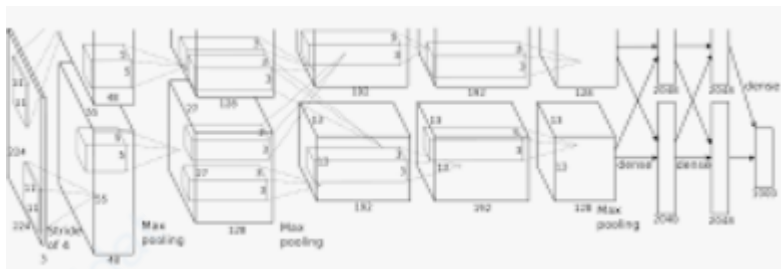
LSTM cell



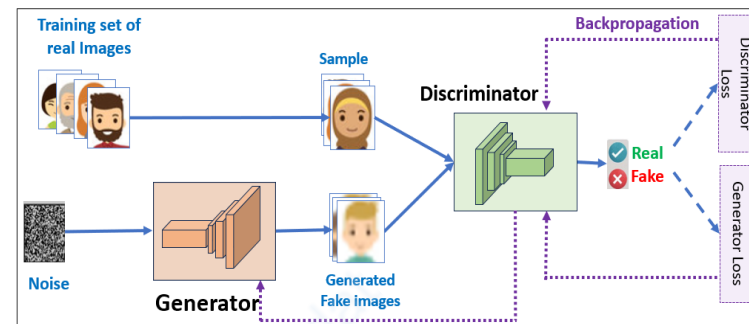
Key Milestones in Deep Learning History

2010s: Breakthroughs and Widespread Adoption

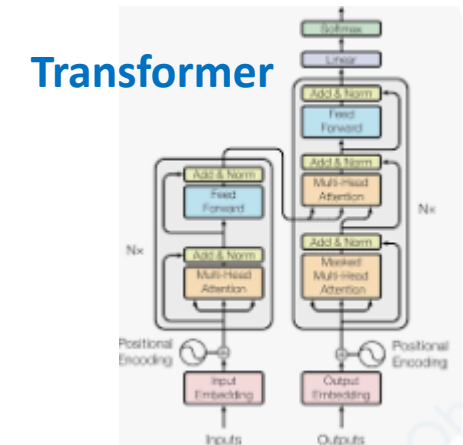
1. 2012: AlexNet, developed by Krizhevsky, Sutskever, and Hinton, won the ImageNet competition, demonstrating deep learning's superiority in image recognition.
2. 2014: Ian Goodfellow introduced **Generative Adversarial Networks (GANs)** for generating synthetic data.
3. 2015: ResNet by He et al. introduced very deep networks with skip connections, improving performance.
4. 2017: Google introduced **Transformer models**, leading to advancements in NLP (e.g., BERT, GPT models).



AlexNet



GANs

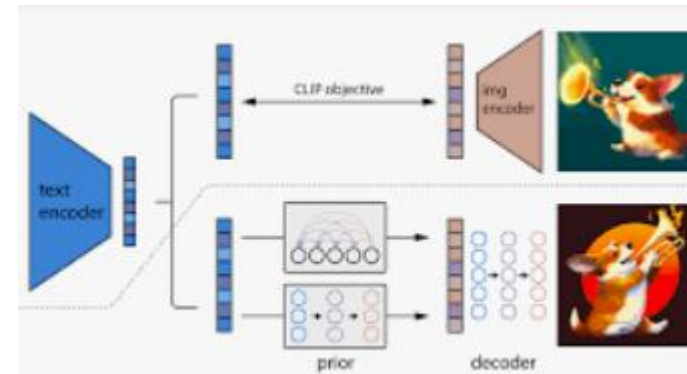


Transformer

Key Milestones in Deep Learning History

2020s-Present: AI Revolution and Large-Scale Models

1. 2020: OpenAI released **GPT-3**, a large-scale language model with human-like text generation capabilities.
2. 2021: DeepMind's **AlphaFold** solved a 50-year-old problem in protein folding.
3. 2022-Present: Models like **ChatGPT**, **DALL·E**, and **Stable Diffusion** showcase deep learning's creative potential.



diffusion model for image generation

